

BUGULMA, Russian Federation

WMO: 287110

Lat: 54.6329N

Lon: 52.8004E

Elev: 304

StdP: 97.73

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-24.0	-30.5	0.2	-27.2	-27.1	0.3	-23.3	13.1	-6.2	11.9	-6.2	2.6	300	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	30.3	19.1	28.4	18.6	26.8	17.6	20.6	27.4	19.7	26.5	18.9	25.0	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.6	23.7	17.4	12.9	22.8	16.5	12.2	21.9	60.7	27.3	57.7	26.7	54.7	25.1	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.4	8.3	-29.9	32.6	3.9	1.9	-32.8	33.9	-35.1	35.1	-37.3	36.1	-40.1	37.5		
(5)			-30.1	21.9	3.9	1.2	-32.9	22.8	-35.2	23.5	-37.3	24.2	-40.2	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-11.7	-11.2	-5.0	5.1	12.8	17.2	19.1	17.5	11.6	4.3	-3.9	-9.8	(6)
(7)		DBStd	12.49	6.43	6.46	4.61	6.12	5.33	4.43	3.94	4.42	4.79	4.93	5.75	6.60	(7)
(8)		HDD10.0	3193	672	592	464	173	33	3	0	1	36	189	416	613	(8)
(9)		HDD18.3	5424	930	826	722	398	184	72	40	70	208	437	666	871	(9)
(10)		CDD10.0	976	0	0	0	26	119	221	281	233	84	11	0	0	(10)
(11)		CDD18.3	165	0	0	0	1	12	39	63	44	6	0	0	0	(11)
(12)		CDH23.3	1456	0	0	0	7	115	342	553	395	44	0	0	0	(12)
(13)		CDH26.7	400	0	0	0	1	15	91	158	130	6	0	0	0	(13)
(14)	Wind	WSAvg	3.9	4.4	4.4	4.7	4.1	4.0	3.3	2.8	3.0	3.3	4.2	4.3	4.4	(14)
(15)	Precipitation	PrecAvg	516	26	22	26	30	46	72	59	50	49	53	43	33	(15)
(16)		PrecMax	778	58	40	52	68	166	178	147	107	113	119	91	62	(16)
(17)		PrecMin	302	3	1	5	1	5	5	1	18	8	9	5	4	(17)
(18)		PrecStd	106	13	11	13	20	35	41	37	25	29	28	23	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.0	7.0	24.1	28.4	31.6	32.8	33.2	27.3	19.0	9.4	1.6	(19)
(20)			MCWB	0.0	0.0	3.5	12.9	16.2	18.8	19.6	19.0	14.8	11.0	8.0	0.6	(20)
(21)		2%	DB	-0.3	0.1	4.0	20.5	26.5	29.2	30.1	29.9	24.2	16.5	6.3	0.6	(21)
(22)			MCWB	-0.8	-0.4	1.4	11.5	15.1	18.4	19.9	18.7	14.5	10.0	5.3	-0.1	(22)
(23)		5%	DB	-1.4	-0.9	2.3	17.8	24.3	27.1	28.3	27.6	21.6	13.8	4.5	-0.2	(23)
(24)			MCWB	-1.9	-1.6	0.6	10.0	14.6	17.7	19.4	18.3	13.8	8.9	3.5	-0.8	(24)
(25)		10%	DB	-2.9	-2.4	1.1	14.7	21.9	25.0	26.6	25.1	19.0	11.4	2.8	-1.3	(25)
(26)			MCWB	-3.5	-3.1	-0.2	8.5	13.3	17.2	18.7	17.2	12.8	7.8	1.9	-1.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.2	4.5	13.8	18.0	21.0	21.9	20.8	17.1	12.8	8.2	1.0	(27)
(28)			MCD B	0.5	0.5	6.9	22.5	25.2	27.9	29.0	27.9	23.5	16.0	9.2	1.1	(28)
(29)		2%	WB	-0.7	-0.4	1.8	12.1	16.2	19.9	21.0	19.7	15.7	11.0	5.5	0.1	(29)
(30)			MCD B	-0.4	0.0	3.3	18.7	24.0	27.0	27.6	27.5	21.3	14.8	6.2	0.5	(30)
(31)		5%	WB	-1.9	-1.5	0.9	10.5	15.2	18.8	20.2	18.8	14.5	9.3	3.6	-0.8	(31)
(32)			MCD B	-1.4	-0.9	2.0	17.2	22.6	25.0	26.6	25.9	19.4	12.8	4.3	-0.3	(32)
(33)		10%	WB	-3.5	-3.1	0.0	8.7	13.9	17.8	19.3	17.8	13.4	8.1	2.1	-1.9	(33)
(34)			MCD B	-2.9	-2.5	1.0	14.2	20.9	23.5	25.3	24.0	18.0	11.0	2.7	-1.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	6.3	8.1	10.1	9.7	9.8	9.3	8.2	5.9	4.4	5.1	(35)
(36)			MCDBR	5.2	5.4	6.0	11.9	12.5	12.5	12.1	12.4	11.5	9.7	5.0	4.5	(36)
(37)			MCWBR	5.2	5.2	4.5	5.9	5.5	5.6	5.2	4.8	5.5	5.5	4.5	4.5	(37)
(38)		5% WB	MCDBR	5.3	5.1	5.1	11.7	11.6	10.9	11.0	11.2	10.3	8.5	4.7	4.4	(38)
(39)			MCWBR	5.3	5.1	4.2	6.4	5.9	5.5	5.2	5.0	5.6	5.5	4.5	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.310	0.358	0.404	0.389	0.385	0.398	0.421	0.371	0.344	0.310	0.282	(40)	
(41)		taud	2.063	2.056	1.988	2.112	2.298	2.355	2.317	2.234	2.350	2.321	2.205	2.041	(41)	
(42)		Ebn at Noon	620	747	791	814	855	862	841	792	788	717	599	526	(42)	
(43)		Edh at Noon	69	99	134	137	121	115	118	121	94	77	60	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.85	1.91	3.32	4.40	5.43	5.89	5.82	4.61	2.96	1.39	0.59	0.50	(44)	
(45)		RadStd	0.11	0.18	0.36	0.44	0.44	0.52	0.64	0.53	0.42	0.39	0.33	0.13	0.12	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page